

11 Show the happy face

Next, create a function called `show_happy()` to reveal the happy expression when you move the mouse-pointer over Screen Pet as if you were stroking it. Type this code beneath the `blink()` function you added in Step 8.

The `if` line checks to see if the mouse-pointer is over the pet.

```
root.after(3000, blink)
```

```
def show_happy(event):
```

```
    if (20 <= event.x <= 350) and (20 <= event.y <= 350):
```

```
        c.itemconfigure(cheek_left, state=NORMAL)
```

```
        c.itemconfigure(cheek_right, state=NORMAL)
```

```
        c.itemconfigure(mouth_happy, state=NORMAL)
```

```
        c.itemconfigure(mouth_normal, state=HIDDEN)
```

```
        c.itemconfigure(mouth_sad, state=HIDDEN)
```

```
    return
```

`event.x` and `event.y` are the coordinates of the mouse-pointer.

Show the pink cheeks.

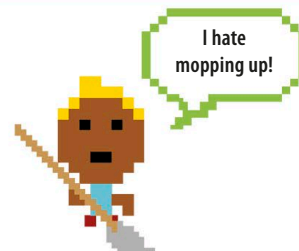
Show the happy mouth.

Hide the normal mouth.

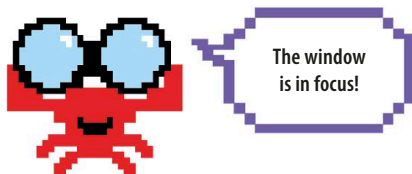
Hide the sad mouth.

LINGO**Event handler**

The function `show_happy()` is an event handler. This means it's only called when a particular event happens, so that it can deal with it. In your code, stroking your pet calls `show_happy()`. In real life, you might call a "mop the floor" function to handle a "spill drink" event!

**EXPERT TIPS****Focus**

Tkinter won't be able to spot you moving the mouse-pointer over the window to stroke Screen Pet unless the window is "in focus". You can get it in focus by clicking once anywhere in the window.

**12 Happy moves**

When the program starts, Screen Pet blinks without you doing anything. But to get it to look happy when it's being stroked, you need to tell it what event to look out for. **Tkinter** calls the mouse-pointer moving over its window a `<Motion>` event. You need to link this to the handler function by using **Tkinter's** `bind()` command. Add this line to the main part of your program. Then run the code and stroke the pet to try it out.

```
c.pack()
```

```
c.bind('<Motion>', show_happy)
```

This command links the moving mouse-pointer to the happy face.

```
root.after(1000, blink)
```

```
root.mainloop()
```